Now, the focus is on deploying the ComicCraft application on a local system using FastAPI. This involves configuring the server environment, installing all required dependencies, setting up static files and templates, and launching the app locally to simulate a real-world environment.
This milestone verifies that the complete comic generation workflow — including AI text generation, image creation, layout building, and PDF export — runs smoothly before planning deployment to a cloud platform in the future.

Activity 5.1: Preparing the Application for Local Deployment

Set up a virtual environment
Begin by creating a virtual environment to manage dependencies and ensure the deployment is isolated from other Python projects. Activate the environment and install all required libraries listed in the requirements.txt file, including FastAPI, Uvicorn, Google Generative AI, Hugging Face Diffusers, FPDF, and others.

```
python -m venv env
# For Windows
env\Scripts\activate
# For macOS/Linux
source env/bin/activate

pip install -r requirements.txt
```

Configure environment variables

- Use environment variables to store sensitive data such as the Gemini API Key and Hugging Face API token
- These variables can be set directly in the terminal or added to a .env file which is loaded into the application using python-dotenv.
  GEMINI_API_KEY=your_gemini_api_key_here
  HF_API_KEY=your-huggingface-api-key-here

Frontend-Backend Rendering via Jinja2

All user interactions are facilitated via **Jinja2**-rendered templates:

- index.html: Collects user inputs
- result.html: Displays the generated AI plan
- all_users.html: Lists all user records for admin view

Activity 5.2: Testing and Verifying Local Deployment

Start the FASTAPI Server

Navigate to the project root where your app/main.py file is located.
Use Uvicorn to launch the FastAPI development server with hot reload enabled:

uvicorn app.main:app --reload

Access the Application Locally

Once the server is running, open any web browser and navigate to:

http://127.0.0.1:8000

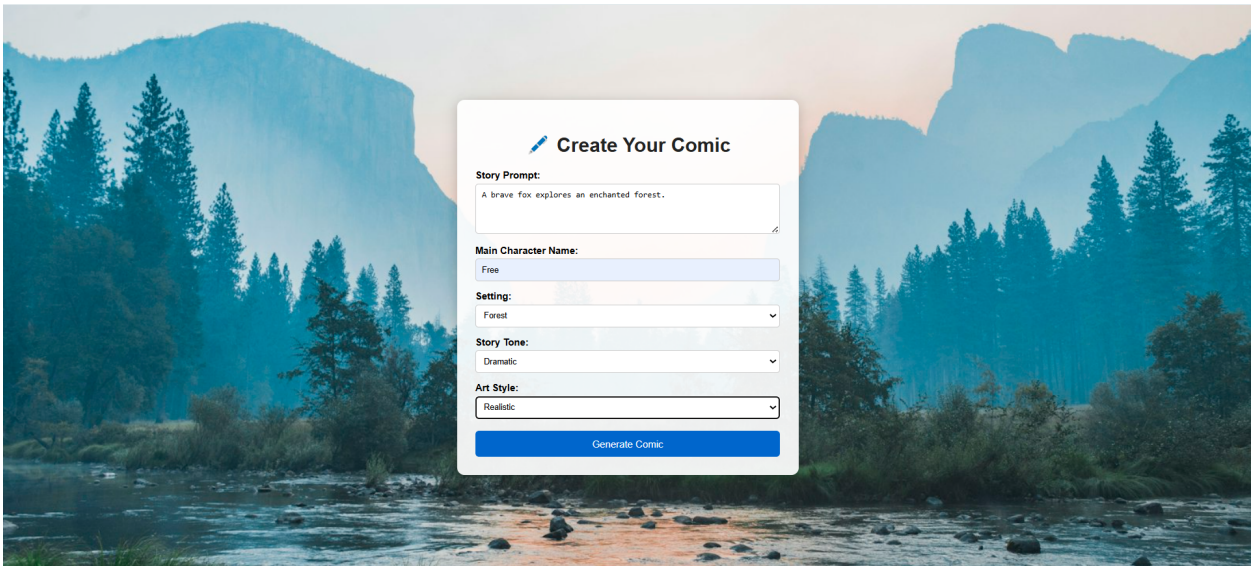
This URL loads the ComicCraft homepage, where users can input their story prompt, character name, setting, tone, and art style to generate a personalized AI-powered comic.

Additionally, the interactive API documentation provided by FastAPI can be accessed at:

http://127.0.0.1:8000/docs
This documentation allows developers to test available API endpoints, such as /generate-comic/json, /generate, /export-success, and /test-image.

Exploring Website's web pages:

Home page:



Description:

The homepage of ComicCraft is designed as an intuitive starting point for users to create AI-generated comic stories. It features a clean layout with a scenic background that fosters a creative atmosphere.

At the center, a structured form collects user inputs necessary for comic generation. Users can:

- Enter a Story Prompt describing their comic idea,
- Specify a Main Character Name,
- Choose a Setting (school, forest, space, city),
- Select a Story Tone (light-hearted, dramatic, poetic, funny),
- Pick an Art Style (anime, pixel art, comic book, realistic).

Once the form is submitted, the system uses AI models to automatically generate a complete comic, including storyline outlines, detailed dialogues, illustrations, and an exportable PDF.

The homepage is designed to be simple, user-friendly, and visually appealing, encouraging creativity and making the comic creation process accessible to all users without requiring technical expertise.

Comic Preview- Result page:

